S5 Table.

Prior work and how this study relates to it

Supporting information for "Environmental fluctuation strengthens gene regulation only when it can be sensed"

The table sets out which parts of the argument are established in the literature, which conflict with our results, and which we did not find precedent for. Literature searches covered theory of plasticity and cue reliability, gene-regulatory-network evolution under fluctuating selection, digital evolution (Avida, Aevol), network sparseness and modularity, and weighted network measures.

Claim	Status	Sources and relation to this study
Reliable cues favour plastic responses; without them, fluctuation favours generalists, bet-hedging or genetic tracking	Established	Levins 1968 [1]; Moran 1992 [2]; Kussell and Leibler 2005 [3]; Botero et al. 2015 [4]. Our results concern the corresponding effect on the regulatory network, not the phenotype.
In a gene regulatory network model, plasticity evolves mainly under fast, erratic change with reliable cues	Established	Jorritsma and van den Berg 2026 [5]. Their abstract reports which parts of the network become sensitive to mutation. We have not been able to read the full text; if it reports connection strength by cue reliability, part of our architectural result would overlap with it.
Fluctuating environments can select for or against gene regulation	Established experimentally	Pollack, Nozoe and Kussell 2025 [6]. There the deciding factor is metabolic flux, not cue information.
Cue-free fluctuation promotes complex regulatory networks	Published; conflicts with our R-Aevol results	Tsuda and Kawata 2010 [7] (optimum moving by a random walk across generations, constant uninformative input). Our grain experiment tested whether grain reconciles the two and found that it does not. Remaining candidates: novel optima versus two alternating targets, and differences in the genotype–phenotype map.
Fluctuating environments favour larger genomes	Published; matches earlier EvoSim results, not R-Aevol	Bentkowski et al. 2015 [8].
Genome size is limited by selection for mutational robustness and falls with population size and mutation rate	Established in Aevol	Knibbe et al. 2007 [9]; Luiselli et al. 2024 [10]. Consistent with the small genomes in large EvoSim populations.
Sensing and non-sensing populations evolve differently in fluctuating environments	Published, different measures	Lalejini et al. 2021 [11] (Avida): trait loss and accumulation of deleterious mutations, not network architecture.
Connectivity is measured by counting edges	Common practice	Tsuda and Kawata 2010 [7]; Kashtan and Alon 2005 [12]; Espinosa-Soto 2018 [13] (binary edges, weights not discussed).
Edge counts can grow by near-neutral accumulation	Established	Soyer and Bonhoeffer 2006 [14].
Summed edge strength as a network measure	Standard method	Iwasaki, Tsuda and Kawata 2013 [15].
The yoked control	Established in the study of learning	Church 1964 [16] describes a bias that arises when individual subjects are paired; our arms are not paired, so it does not apply.

Not found in the literature

Claim	Status	Sources and relation to this study
Edge count and edge strength change in opposite directions under fluctuation	This study	EvoSim (exploratory), R-Aevol informative-cue populations (pre-registered secondary analysis).
A yoked control that separates a cue's information from its presence in the evolution of regulatory networks	This study	R-Aevol, with switching rates matched between arms (S2 Text).
Regulatory strength and genome size compared across informative, yoked and absent cues	This study	Pre-registered predictions P1–P4; subject to the caveat on [5] above.

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